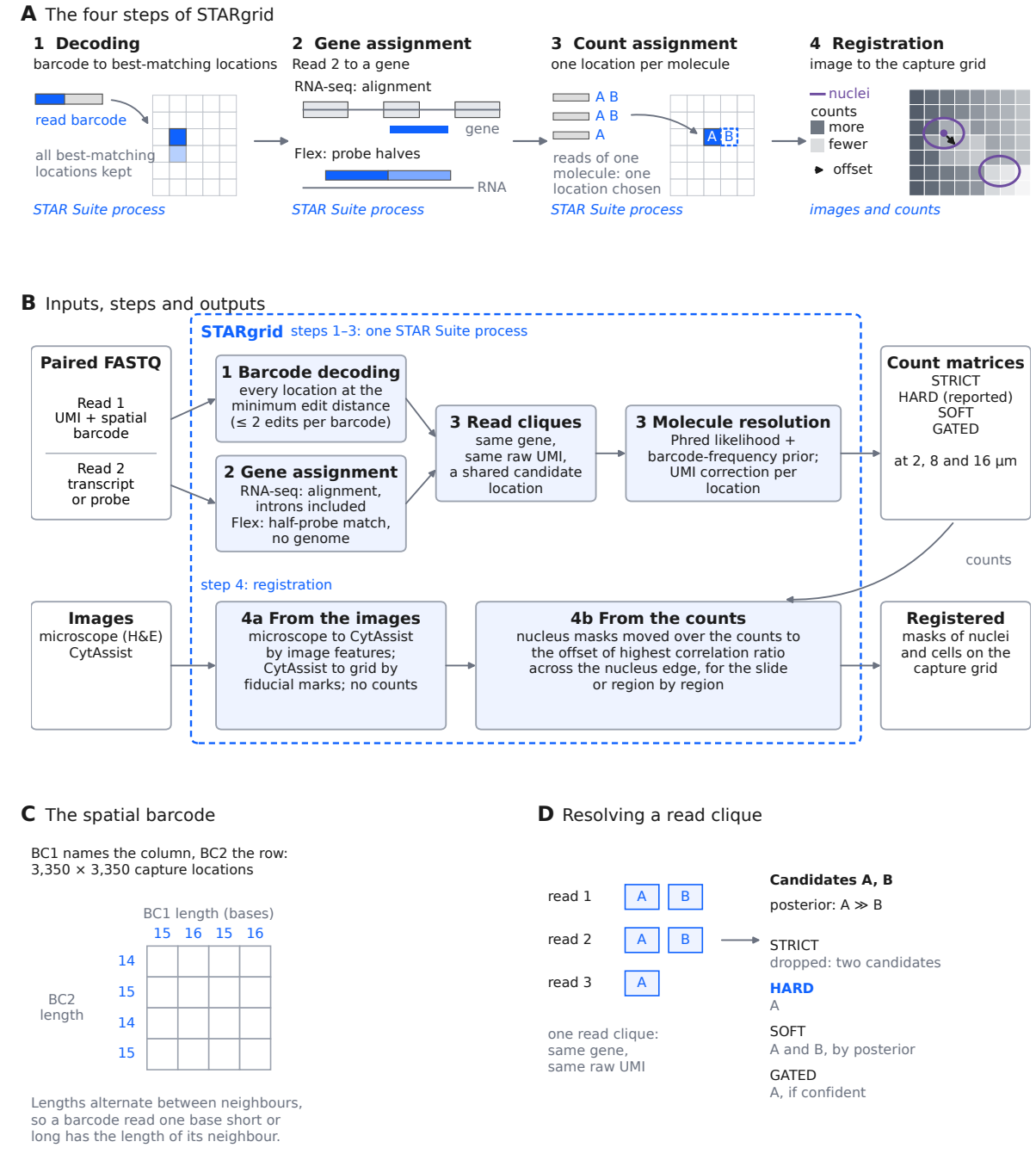
A The four steps of STARgrid
1 Decoding
barcode to best-matching locations
read barcode
all best-matching locations kept
STAR Suite process
2 Gene assignment
Read 2 to a gene
RNA-seq: alignment
gene
Flex: probe halves
RNA
STAR Suite process
3 Count assignment
one location per molecule
A B
A B
A
reads of one molecule: one location chosen
STAR Suite process
4 Registration
image to the capture grid
nuclei
counts
more
fewer
offset
images and counts
B Inputs, steps and outputs
STARgrid steps 1–3: one STAR Suite process
Paired FASTQ
Read 1
UMI + spatial barcode
Read 2
transcript or probe
1 Barcode decoding
every location at the minimum edit distance (≤ 2 edits per barcode)
2 Gene assignment
RNA-seq: alignment, introns included
Flex: half-probe match, no genome
3 Read cliques
same gene, same raw UMI, a shared candidate location
3 Molecule resolution
Phred likelihood + barcode-frequency prior; UMI correction per location
Count matrices
STRICT
HARD (reported)
SOFT
GATED
at 2, 8 and 16 µm
counts
step 4: registration
Images
microscope (H&E)
CytAssist
4a From the images
microscope to CytAssist by image features; CytAssist to grid by fiducial marks; no counts
4b From the counts
nucleus masks moved over the counts to the offset of highest correlation ratio across the nucleus edge, for the slide or region by region
Registered
masks of nuclei and cells on the capture grid
C The spatial barcode
BC1 names the column, BC2 the row:
3,350 × 3,350 capture locations
BC1 length (bases)
15 16 15 16
BC2 length
14
15
14
15
Lengths alternate between neighbours, so a barcode read one base short or long has the length of its neighbour.
D Resolving a read clique
read 1
read 2
read 3
A
B
one read clique:
same gene,
same raw UMI
Candidates A, B
posterior: A ≫ B
STRICT
dropped: two candidates
HARD
A
SOFT
A and B, by posterior
GATED
A, if confident